www.csvtuonline.com

Roll No. .....

# 324353(25)

## B. E. (Third Semester) EXAMINATION, April-May, 2015

## (New Course)

## (Branch : Elect.)

## BASIC ELECTRONICS

*Time : Three Hours* ] [ *Maximum Marks : 80*

[*Minimum Pass Marks : 28*

**Note :** (i) Part (a) of each question is compulsory.

(ii) Attempt any *two* Parts from (b), (c) and (d).

1. (a) What do you mean by law of junction ? State and give expression with notations. 2

(b) A sample of Ge is doped to the extent of $10^{14}$ donor atoms/cm$^3$ and $7 \times 10^{13}$ acceptor atom/cm$^3$. At the temperature of sample the resistivity of pure (intrinsic) Ge is 60 Ω-cm. If the applied electric field is 2 V/cm, find the conduction current density. 7

(c) Derive the following expression for a step graded junction : 7

$$V_o = V_T \ln (N_A N_D / n_i^2)$$



www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com



(d) Draw and explain the V-I characteristics of a junction diode in both forward and reverse biased. 7

2. (a) Define ripple factor. Find the ripple factor for H. W. R and F. W. R. 2

(b) Explain construction and working of bridge rectifier circuit and give comparison between full wave center-tap and bridge rectifier circuit. Why bridge rectifier prefer over center-tap full wave rectifier ? 7

(c) Derive the expression for ripple factor for full wave rectifier with LC filter and also prove that : 7

$$L_C = \frac{R_L}{3\omega}$$

(d) (i) The avalanche diode regulates at 50 V over a range of diode current from 5 to 40 mA. The supply voltage V is 200 V. Calculate R to allow voltage regulation from a load current $I_L = 0$ upto $I_{max}$, the maximum possible value of $I_L$. What is $I_{max}$ ?

(ii) If R is set as in part (i) and the load current is set at 25 mA. What are the limits between which V may vary without loss of regulation in the circuit ? 7

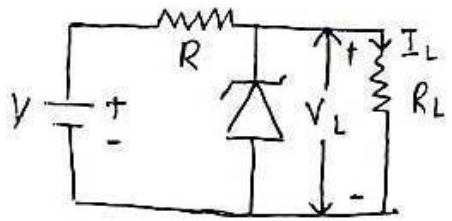


Fig. 1



www.csvtuonline.com

www.csvtuonline.com



3. (a) Why is emitter heavily doped and base is lightly doped in transistor ? 2

(b) (i) For the transistor used in the circuit $h_{FE} = 100$ determine whether or not the silicon transistor is in saturation and find $I_B$ and $I_C$.

(ii) Repeat part (i) with 2 K emitter resistance added, neglecting $I_{CO}$. 7

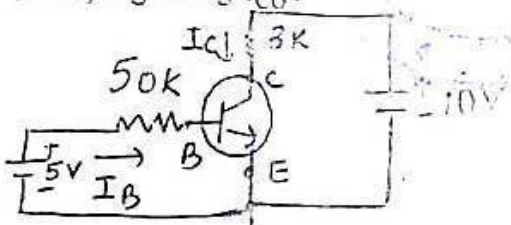


Fig. 2

(c) Explain current component for P-N-P transistor. 7

(d) Compare CE, CB and C. C. configurations. Explain why CC configuration is called emitter follower. 7

4. (a) Define transconductance ($g_m$) and drain resistance ($r_d$) of an F. E. T. 2

(b) For an $n$-channel FET having $I_{DSS} = 1$ mA, $V_P = -1$V. If the quiescent drain to ground voltage is 10 V, find $R_1$. 7

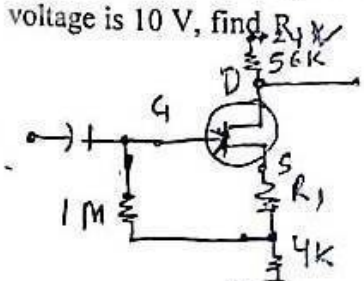


Fig. 3



www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com



(c) Explain depletion MOSFET w. r. t. the following points :

(i) Construction

(ii) Operation

(iii) Circuit symbol

(iv) Drain and transfer characteristics

(v) Advantages

(vi) Application

(d) (i) Write any *three* differences between [illegible] and JFET.

(ii) For an $n$-channel JFET $I_{DSS} = 8.7$ m[illegible] $V_P = -3$V and $V_{GS} = -1$ V. Find [illegible] value of $I_D$, $g_{mo}$ and $g_m$.

5. (a) Draw the RC low pass and high pass circuits [illegible] find the transfer function.

(b) If a sine wave of 5V peak to peak at 1 kHz [illegible] applied to series and shunt clipper circuit [illegible] the output waveforms. Assume d. c. supply of 2[illegible] and diode is ideal.

(c) Explain crystal oscillator. Also write [illegible] advantage, disadvantage and application [illegible] crystal oscillator.

(d) Describe the R-C phase shift oscillator usi[illegible] transistor.

400

www.csvtuonline.com

www.csvtuonline.com